

Second Edition

RESEARCH METHODOLOGY IN APPLIED ECONOMICS

Organizing, Planning, and Conducting
Economic Research

Don Ethridge

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Philosophical Foundations

Economic thinking and research methodology in economics have been and will continue to be influenced by different philosophical positions held by individuals and by groups of individuals with similar views and beliefs. Each philosophical position holds views about and offers prescriptions for what is "proper" in terms of activities, attitudes, and approaches used in economic research. There are many philosophical value positions that might be identified, but we will focus on three of them—positivism, normativism, and pragmatism—that have had a major role in shaping economic thinking and ideas of conducting research in economics. Note that the labels for these philosophies may vary, and suffice it to say that there is considerable disagreement among individual economists, and groups of economists, on these philosophical positions and the relative importance of each in its effect on economic thinking. In this chapter an attempt is made to synthesize or generalize about these philosophical approaches. The intent is not to choose among them or to advocate one over another, but rather to introduce and illustrate how these philosophies have contributed to methodological approaches in economic research.

One of the reasons why we need to understand something about philosophical positions is to be able to deal with the beliefs of others as well as our own attitudes and beliefs. The following, from a personal communication with George W. Ladd, is offered to provide perspective on this point:

There are some aspects of economists' behavior that I could not understand until I studied philosophy of science. I will illustrate with a personal experience with two economists—call them A and B—whom I have always respected and still do. Initially A did not respect B, and I could not understand why, but B respected A. Now each respects the other. B does problem solving or subject matter research (Johnson's terminology); most of it is normative or conditionally normative; much is multidisciplinary. A has always done strictly disciplinary research that is positivistic, and he adheres to Friedman's philosophy. When he was younger, A thought that economists should only do positivistic disciplinary research, and therefore believed that B was not being an economist. B, however, has always recognized that the profession can profit from different philosophies and different kinds of research. As A has matured, he has come to share B's broader view of the profession.

One's philosophical beliefs affect one's choice of "legitimate questions" (i.e., of one's selection of issues that are proper for economists to study) as well as one's choice of research methods.

The importance of methodology, including its dependence on philosophical beliefs, was also stated succinctly by Randall (1993, p. 58) as follows:

Because researchers are the front-line methodologists for their own research programs, it follows that the serious study of methodology and rhetoric is desirable for researchers. Such study is unlikely to yield hard-and-fast rules for research, but is likely to yield a rich lode of insights. . . . Furthermore, it tends to make researchers more introspective about their own work, which is all to the good.

The nuances of philosophical arguments and views are complex and there are rarely, if ever, simple outcomes to the issues. Different philosophies as they relate to science and research represent different views of what constitutes reliable knowledge, or what is proper activity or procedure for establishing reliability. Each of the general philosophical positions addressed here offers views that are well founded, yet each has premises that can be questioned. This chapter does not resolve any of the conflicts or inconsistencies. But as students of research methodology in economics, we need to understand at least the basic elements of the general philosophical positions and disagreements as we attempt to conduct our research activities, and evolve our own philosophical positions. Remember that the boundaries of research activity are not set by any single philosophical position.

The three philosophies most connected to research methodology are positivism, normativism, and pragmatism. In the material that follows, the ideas and concepts underlying these philosophies are introduced. For a more thorough treatment of them, readers are referred to Glenn Johnson's book (1986b, chs. 4-9) and should consult the references he provides. That material represents an extensive and insightful explanation of the three philosophies, and the summary below draws heavily on Johnson's work. The serious student of the philosophical underpinnings of economic methodology, including research methodology, must read far beyond the treatment of the philosophies presented here.

The philosophies presented are philosophies in the context of confirmation, not in the context of discovery.¹ Confirmation is concerned with determining the reliability of knowledge, while discovery is concerned with the finding of new knowledge in the first place. Confirming the reliability of knowledge is the central concern of the methodology of science, and the philosophies of positivism, normativism, and pragmatism hold different positions or views on how that may be done. This does not say that science is not concerned with discovery, but rather that the philosophical positions do not directly address the process of discovery.

The methodology of research is concerned with confirmation, but it also is concerned with discovery, although the prescriptions for achieving it are nonspecific. We understand that discovery embodies creative thinking, for which the techniques are less structured, while we have more highly developed, structured techniques and procedures for confirmation. The act of discovery may require relatively more creativity and intuition than does the act of confirmation. While this book attempts to encompass both confirmation and discovery, the caveat offered here is that the three philosophies discussed in this chapter are concerned primarily with confirmation, not discovery.

As a prelude to the discussions of positivism, normativism, and pragmatism as central philosophies, a cautionary note may be useful. Do not assume, as some may suggest, that science and research are inherently associated with a particular philosophy. There are still some in the sciences, usually in the physical sciences, who believe that positivism is the only valid philosophy underlying the "scientific method." However, McCloskey (1983, p. 482) makes the case that this concept of the scientific method does not describe the sciences as it was once thought, and that purely positivistic models are not necessarily good models for economics anyway. The position taken here is that all sciences and the collective body of research embody elements of each of these philosophies.

POSITIVISM

Positivism as a philosophy adheres to the view that only "factual" knowledge gained through observation (the senses), including measurement, is trustworthy. "Pure" positivism even discounts reasoning and theory as valid for arriving at reliable knowledge. However, the more modern view of positivism embraces the logical extension of the facts. This is called logical positivism, and it became the predominant positivistic philosophy early in the twentieth century; few in the scientific community subscribe to pure positivism today. Logical positivism is the point of reference henceforth in this discussion.

Positivism originated in the physical sciences and some in those disciplines still view it as *the* philosophy of science even though it is declining in importance in the scientific community relative to the other two philosophies. Logical positivism has influenced economics, particularly in the twentieth century, although it has been waning relative to pragmatism, and has never been the dominant philosophy in economics. Logical positivism was to become influential in economics by the 1950s with proponents such as Wassily Leontief, Milton Friedman, and Harry Johnson (see Friedman, 1953; Johnson, 1975; Leontief, 1993). The philosophy had been building among economists since Alfred Marshall's time. Johnson (1986b) points out that John Neville Keynes, John Maynard Keynes's father, had been an early proponent of logical positivism in economics in the 1890s. The logical positivism of economics may be viewed to include the study of society's values as well as descriptive, value-free knowledge, but some in the physical sciences would not regard any knowledge about values as positivistic.

The philosophy of positivism disavows the reliability or scientific validity of both prescriptive knowledge and descriptive knowledge of "real" values. Logical positivists will accept the validity of knowledge about values people hold regarding things or circumstances (e.g., that people believe that cancer is bad) but will not accept the validity of descriptive knowledge about values as real characteristics (e.g., that cancer is bad irrespective of beliefs). That is, positivists accept the "measurement" of people's or society's beliefs about what is good or bad, but not the knowledge that anything is inherently or descriptively good or bad. Concepts of good/bad or right/wrong are viewed by positivists as constructs of the mind, not as describable characteristics of reality. The view of the positivist is that only things that can be directly observed or measured are valid for scientific attention, and things that are not observable or measurable are not meaningful to the positivist; some philosophers of science use the term *standard empiricism* to describe positivism (Maxwell, 1992). In positivism, the emphasis on experience is so encompassing that there is no recognition of the significance of a practical or "real-world" problem as a reason to observe and assemble facts and postulate hypotheses (Back, no date). Lewin (1996, pp. 1305, 1306) states:

According to logical positivism, science does not seek to help us to *understand* the nature of reality; that is the domain of metaphysics. *Understanding* is . . . a meaningless pursuit. The world simply consists of observable empirical regularities, and science should restrict itself to *describing* these, in the form of objective, falsifiable propositions. Every scientific concept and theory should have a clear empirical meaning; no ambiguity can be tolerated. . . . Therefore, if economics is to be a science, then we must transform the tautological propositions of pure theory into falsifiable scientific propositions, by assigning precise empirical meanings to all theoretical concepts.

However, positivism, like all philosophies, is based on empirically untested presuppositions. Positivists reject untested presuppositions that good or bad is empirically knowable but accept the untested presupposition that experience is empirically knowable. This position may "grow out of the failure to distinguish between the good and bad and the right and wrong (which are the logical consequences of decision rules and therefore not part of the natural world to be observed)" (Johnson, 1986b, p. 43). Positivistic knowledge is also culturally dependent because the experience through which we interpret observations is affected by the surrounding culture.

Castle (1989, p. 3) notes that logical positivism holds that theoretical concepts are valid only if the theory or its propositions can be quantified. This position is also too extreme a position for many economists, although we would prefer to include quantification whenever possible. Quantification is not always possible, however. Another point worth noting is that logical positivists tend to see both facts and theory as sources of hypotheses. Economists embrace this component of positivism, seeing both theory and facts as contributing to hypotheses, often by interacting; this is discussed further in chapter 8.

Economists cannot fully embrace the philosophy of positivism. This is largely because many things that are not concrete are nonetheless "real." For example, no one has ever directly observed a demand relationship, but its conceptualization and representation are real and its characteristics can be estimated; it is logically positivistic. One does not necessarily have to be able to touch something for it to be real. A commodity demand relationship is, to us, as real as the physical existence of a chemical element. Because a value must be "seen" through a logical process of conceptualization does not detract from its existence. Some things can be seen only with the aid of an electron microscope; some things can be seen only with the aid of reasoning. Thus, the senses are not the only means through which experience leads to information or knowledge.

While economists do not endorse the philosophy of logical positivism to the exclusion of other philosophies, it has had a profound effect on economic thinking and economic research. One way it has influenced economics is to place more emphasis on measurement and quantification when possible. Empirical analysis and estimation get at least part of their impetus from the thrust to be more positivistic. New methods in statistics and econometrics are influenced by positivistic thinking. However, positivism and empiricism are not synonymous. Empiricism includes measurement of things and phenomena but extends into quantification and estimation of relationships. In fact, Caldwell (1982, pp. 19-35) characterizes logical positivism as evolving toward or maturing into something he characterizes as logical empiricism, which is a sort of merging of logical positivism with pragmatism.

Another major way in which positivism has influenced economics is to focus attention on knowledge about values as being positivistic knowledge, while prescriptive knowledge is substantially different. For example, behavioral attributes of people, institutions, or groups are positivistic when they are quantifiable and/or demonstrable. Identifying, quantifying, or demonstrating what people value, or how much they value things, is little different from positivistic knowledge about physical things that can be observed or measured.

A third way in which positivism as a philosophy has had an effect on economic thinking is to highlight the importance of objectivity in the practice of economics and economic research. That is, positivistic philosophy, in emphasizing the importance of providing evidence to support a descriptive conclusion, argues the nebulousness of personal judgments and perceptions. A paradox of this stance, however, is that judgments are inherent, not just with respect to value, but with respect to "facts" or "reality." Rudner (1953, p. 2) explains that scientists inherently make value judgments about things like confidence intervals on descriptive parameters. When we set confidence limits for statistical tests, we make a decision about what level of evidence is *sufficiently* strong to warrant not rejecting a hypothesis. He proceeds to argue that "objectivity for science lies at least in becoming precise about what value judgements are being and might have been made in a given inquiry" (p. 6). To cast it slightly differently, perhaps the

useful admonishment of positivistic philosophy, and our consideration of it, is to remind us to be on guard to distinguish between private and public knowledge (i.e., between what we accept as valid in our individual private lives and what we can convincingly demonstrate to others using experience and logic).

The tests of logical positivism are the tests of correspondence, coherence, and clarity discussed in chapter 3. These tests fail to recognize, as we shall discuss later, any consideration of usefulness or applicability. This makes the philosophy of logical positivism incomplete in a sense but does not damage the philosophy as a part of a more general approach to research in economics.

The behavior of economic agents and entities and their valuation of things is a legitimate part of positivism, although nonbehavioral aspects are not because positivists believe we cannot measure values as such. Positivistic philosophy affects disciplinary and subject-matter research but does not accept the prescriptive objectives in problem-solving research in economics. Its role in disciplinary research is seen in its effects on empirical testing in the development, refinement, and validation of theory, in disciplinary knowledge, and in the research on new procedures for studying relationships. In subject-matter research, its effects are most obvious in the estimation of parameters and relationships that provide the basis for policy and management decisions, even though positivists would disavow the prescriptions required of the decision process.

Summarizing positivistic philosophy, it has had important and lasting effects on the way we think about science, research, and reliability of knowledge, but is no longer the dominant philosophy of science. It is too limiting to be dominant, especially in economics but also in many other disciplines. The criticism of positivism as a philosophical position may have been most succinctly stated by a physicist (Heisenberg, 1991, p. 826):

To positivists, the world must be divided into that which we can say clearly and the rest, which we must pass over in silence. But it is a pointless philosophy, because we would be left with trivial tautologies.

But irrespective of these criticisms, Johnson (1997, p. 262) makes this case:

Logical positivism has and still does provide philosophic orientation for a considerable part of mainstream determinate disciplinary economics and it continues to guide the work of most physical and biological scientists, even if it is being subjected to increasing criticism from leading intellects.

NORMATIVISM

Normativism as a collection of philosophies (Johnson, 1986b, p. 34) related to science takes the position that knowledge of the goodness and badness of conditions, situations, things, and actions is valid and even necessary in order to produce prescriptive knowledge. Some forms of normativism, such as utilitarianism,

hold that good and bad are experientiable (e.g., cancer *is* bad). Some hold that descriptive knowledge of intrinsic value of good and bad is possible and that this knowledge can be used to draw prescriptive conclusions about what is right and wrong or of what should or should not be done. To normativists, conditions and situations can be good or bad, but they cannot be right or wrong. Actions can be both good or bad and right or wrong. The view presented here is that normativistic philosophical beliefs are more prominent in the broad range of problem-solving and subject-matter economic research than in the more restrictive activity of disciplinary economic research within economics. Normativistic philosophy in economic research emphasizes matters on which people place value—things such as efficiency, welfare, income, standard of living, and quality of life. To the normativist, as the research relates to activities like business and policy decisions and recommendations, goodness and badness considerations become more central to the prescriptive process. Normativists accept the presupposition that intrinsic values (good and bad) are knowable, if only from a given perspective. They accept, however, that good and bad are as they are because people believe them to be.

Normativism as it applies to economic research is not directly concerned with moral questions of or moral prescriptions on right and wrong, which are fundamentally different from questions or knowledge of what is good and bad (Lewis, 1955). However, we often take as given the societal rules that govern societies and the markets that function within them, which are essentially moral/ethical rules (Johnson, 1986a, pp. 73, 74). Economic choices may, perhaps often, entail selecting between several "bad" alternatives; the choice of a "right" alternative under the circumstances may not involve a "good" choice or outcome. Whether a condition, situation, or action is right or wrong in a moral sense may not be a concern within economics. Thus, when we talk of normativistic philosophy as it relates to economics and to economic research, we do not encompass *all* normativistic philosophy. The purpose is to avoid the incursion of personal private values and maintain the focus on public rather than private knowledge of both intrinsic and extrinsic values. This, consequently, places emphasis and importance on the objectivity of researchers.

Objective normativism² (Johnson, 1986b, pp. 57–62) refers to a position that the desirability (goodness) of a result or outcome (or lack of it) from the point of view of individuals or groups can be known (e.g., based on observation or experience and logic), whatever the beliefs are and for whatever reason they may exist. Situations, actions, and so on can then be analyzed in terms of their goodness/badness given those views (values). Objective normativism as a philosophy is not very different from the philosophical position of logical positivism. Johnson (1986b, pp. 59–60) states:

With respect to value-free positive knowledge, it is important to recall that methods used in acquiring such knowledge involve the subjectivity of sense impressions, the necessity of interpreting sense impressions, leaps of faith

from sense impressions to reality, insight, intuition, and the need for interpersonal acceptance of perceptions of reality. This leads to the position that objective descriptive knowledge about the value of conditions, situations, and things is fundamentally similar to knowledge on the value-free positivistic side, and that both value-free positivistic and value propositions are judgmental in nature. We also note that on both sides we have to put up with both probability distributions and the possibility of making mistakes in interpreting our sense impressions and in employing our logic. Further, it should be kept strictly in mind that we are only considering knowledge of values here and are excluding prescriptive knowledge of rightness or wrongness.

Objective normativism accepts that objective value knowledge is sometimes essential for statements or prescriptions about what should be done to accomplish specified goals or objectives. For example, "The people of the world are better off with free trade than with restricted trade" is an objective normative statement. The statement, "If the goal is to increase farmers' incomes with minimum impact on the functioning of commodity markets, a program of direct income supports rather than one of commodity price supports constitutes the better policy" is a conditionally normative statement that also qualifies as objectively normative (assuming it is based on sound analysis) because the assumption (condition) is explicit. If the statement becomes, "If the goal is to increase farmers' incomes with minimum impact on commodity markets, the government should adopt a program of direct income supports" it is objectively normative *and* prescriptive.

Normativistic philosophy is an inherent part of economics and economic research, and there is no reason to consider it inferior or to apologize for it—as long as it is kept objective as contrasted with subjective (which is necessary for both value-free and value knowledge). In fact, it is the element of objective normativism that makes the discipline, and our research processes and results, relevant to most of our clientele. These include policy makers of various types, with different concerns and problems, and firm managers and industry leaders. Our clientele also include the general public with an array of concerns for issues ranging from environment and living standards to international trade and national defense. In fact, any discipline that deals with public policy *must* use normative value judgments, and economists/economics must, by definition, be concerned with values, both monetary and nonmonetary. We believe that economics and economic research have something to offer everyone on matters that concern them. That capability would be limited without elements of normativism embodied in the attention of economists. What we must remember is that we are not in a position to impose individually subjective perceptions of value on others.

Objective normativism, as applied in economics, contributes to all three types of knowledge. Its connection with knowledge of values and prescriptive knowledge is direct and obvious. The connection with positivistic knowledge is indirect—through knowledge of values—and is dependent on one's acceptance of

the argument that objective knowledge about values is no different from value-free positivistic knowledge. This philosophy influences disciplinary research through its focus on knowledge about values and behavior. It is exhibited in subject-matter research through both knowledge of values and prescriptive knowledge. Problem-solving research depends more heavily on the prescriptive knowledge emphasis of objective normativism.

PRAGMATISM

Pragmatism as a philosophy holds that what is important with respect to descriptive knowledge is how well it works for solving a problem at hand. Pragmatists are interested primarily, if not completely, in prescriptive knowledge. Attention to theoretical logic for its own sake and distinctions between abstractions such as positivistic knowledge and knowledge about values are relatively unimportant for pragmatists. They view positivistic value-free knowledge and normativistic value knowledge as being interdependent and place little confidence in the distinctions between them or even believe such distinctions are impossible. Pragmatists also may see means of obtaining knowledge and the ends producing the knowledge as interdependent. For example, a prescriptive result of a research process may produce an understanding of consequences that leads to a shift in the desired result. Pragmatists evaluate concepts for their usefulness in solving contemporary problems rather than for their own sake. That is, pragmatists are interested in *applying* concepts to solve problems. As part of this focus on solving problems, pragmatists regard value-free knowledge and knowledge of value as interdependent and inseparable. Workability (appropriateness for the problem at hand) is a central pragmatic criterion for judging empirical propositions (Johnson, 1986b, p. 37).

An example may be useful to help understand the pragmatic interdependence of value-free and knowledge of values. Consider the hedonic price analysis studies done on automobiles, with the objective to determine the market values of attributes or amenities of automobiles rather than the values of automobiles as a composite of attributes (see Griliches, 1961, for example). These studies used data on the attributes of automobiles (e.g., colors, engine types and sizes, existence of air conditioning, and many other attributes), which consisted of value-free knowledge, and data on prices of the automobiles (knowledge of values) to impute the values of the attributes embodied in the autos. Distinguishing between the two types of knowledge was immaterial, and the interdependence was necessary to produce results. The results provided prescription for adjusting price indices for quality change that occurs through time.

Pragmatism as a philosophy affecting economics became a prominent force with the institutionalists in the 1920s. While some assert that the institutionalist school of thought has had little important influence on formal, theoretical economic thought, the matter is certainly debatable. Pragmatism has had a substantial, and perhaps increasing, influence on research methodology in economics. Institu-

tionalists emphasized the importance of society's institutions in economic outcomes, and economists today are more attuned to those roles, as evidenced by the incidence of Nobel Laureates for Economics going to students of public choice and institutional economics. The lasting influences of the pragmatic institutionalist perspective has been to focus economics more on solving problems and to make the economics of institutional change part of the mainstream of economics.

From the pragmatist's perspective, the test of workability is the primary test for reliability of prescriptive knowledge. If a concept works prescriptively (solves the problem) then it is acceptable. However, tests of clarity, coherence, and correspondence also have a role in pragmatism even though the test *may* not be applied before the prescription is implemented. Clarity is achieved if prescriptive solutions to problems are not ambiguous or vague. If the prescription is valid (it works), coherence (absence of contradiction) is achieved (Johnson, 1986b). If the outcome is consistent with what we already know, it meets the test of correspondence.

Pragmatic philosophy has affected economic thinking and research in profound ways, especially in subject-matter and problem-solving research, and especially in the United States during the twentieth century. Possibly the most significant influence has been to place the focus on nondisciplinary problems/issues as reasons to initiate research inquiry. Northrop (1959) attributes this pragmatic focus for research primarily to John Dewey, an early-twentieth-century philosopher. Within economics, pragmatism is a more prominent philosophy with some individuals, groups, and institutions than with others. Generally, pragmatic philosophy plays a larger role in problem-solving research than in subject-matter research, and has the least influence in disciplinary research. Even so, pragmatism plays a major role in most aspects of economic research because much of our focus is on either management or policy issues.

All "decision sciences" embrace pragmatism in some form or to some degree. U.S. educational and political systems are dominated by a philosophy of pragmatism. We are conditioned to be concerned with problem solving and workability (achieving results) in our educational system and through political activities and processes. Most Americans take pride in pronouncing themselves pragmatists; we economists from this culture cannot avoid being influenced by that conditioning. All our research that has potential policy or management applications draws on a pragmatic interest in applicable results/findings. The institutional school of economic thought, the only school identified as distinctly American, is based largely on pragmatism. However, while pragmatism as a philosophy has had a major influence on economics, its effect has been partial and selective, as has been the case with the other two major philosophies. As with the other philosophy positions, pragmatists make their own presuppositions. For example, they accept the presupposition that the difference between knowledge of values (both intrinsic and extrinsic) and value-free knowledge does not matter. They also accept the presupposition that *why* prescriptive knowledge is valid is secondary to concerns that it is valid.

HOW THE PHILOSOPHIES BLEND

Most disciplines and fields of study have adopted and adapted the basic philosophies of logical positivism, objective normativism, and pragmatism in some combination. There are also differences in individuals' philosophies within a given discipline or field of study. For example, mathematicians who are interested in the problem-solving applications of mathematics philosophically embrace pragmatism and/or objective normativism to some extent. But many mathematicians have a low tolerance for pragmatism, certainly compared to most economists. While recognizing these inherent differences, it is nevertheless useful to generalize to some extent about how economics and economists blend the various philosophies in conducting research.

Economics is a problem-solving, decision-oriented discipline by the nature of its focus. The problems/issues may be disciplinary, subject-matter, or problem-solving in nature. Economics addresses all sorts of problems, ranging from contemporary, short-term problems of everyday activity to complex theoretical disciplinary problems of intricate cause-effect relationships. It is the study of human activities in the allocation of limited resources in satisfying human wants; its focus is in allocating these resources among competing uses. Economic research is expected to be relevant to understanding and dealing with current and future societal problems. This expectation exists both within the discipline and in the minds of our external clientele. How well we fulfill those expectations is a matter on which there will be further discussion.

The three philosophies—logical positivism, objective normativism, and pragmatism—may be relatively more or less important according to the type of research. In disciplinary research, in which we can include basic theoretical research, basic descriptive (including measurement) research, and research on specific analytical procedures that may be applied within the discipline, both logical positivism and objective normativism provide philosophical focus. The part of objective normativism that emphasizes knowledge of values as they relate to behavior is particularly important in theoretical constructions that describe and explain economic behavior of individuals and groups. Logical positivism is an important component of the development or adaptation of basic analytical procedures, but pragmatism is certainly not absent in disciplinary research because often disciplinary research is undertaken with a view toward a social problem. However, problem-solving pragmatic prescriptions are usually less prominent in strict disciplinary research than in subject-matter or problem-solving research.

Subject-matter research pulls from all three philosophies, but in ways different than disciplinary research. In subject-matter areas, for example, environmental economics, labor economics, managerial economics, and natural resource economics (there is no definitive list of subject-matter areas), the objectively normative content often leans more toward development of prescriptive knowledge based on both value and value-free knowledge. The philosophical intent is to

develop understanding of outcomes of proposed actions, given specified conditions, in the area of interest. This understanding leads to guidelines and decisions. In any event, prescriptive subject-matter knowledge may be based in part on knowledge of behavior developed through disciplinary research. Subject-matter research also has a pragmatic focus because the prescriptions must achieve the desired effects (they must work). Subject-matter research can be restricted to being only positivistic in terms of its empirical estimation processes and procedures. For example, empirical estimation of the magnitude of impact of pollution user charges on the emission of pollutants or estimation of the amount of an industry's production response to price controls is positivistic.

Problem-solving research that addresses a particular management or policy question for a specific decision maker also can be both positivistic (e.g., in estimating parameters) and normativistic (e.g., in estimating consumer surplus) in its empirical estimation procedures. However, it is conditionally prescriptive (normative) in terms of its objectives (it intends to lead to a decision on what is to be done). Also, it always has pragmatic content. Pragmatism is paramount in problem-solving research because of its emphasis on a particular kind of prescriptive knowledge. While problem-solving research in economics necessarily must pull from conceptually sound and logically applicable theory and employ appropriate analytical techniques, meeting those criteria is not enough; it must achieve the problem-solving objectives of the research. In this regard, Johnson (1986a, pp. 83-84, 86) states:

Many practicing economists have simply ignored the restrictions of these [philosophical distinctions] as impractical abstractions generated by academic economists out of touch with important problems of the real world. . . . We should note that the complex, sometimes clumsy, methods of pragmatism at least address welfare and value questions in an objective manner, which is more than can be said for positivism and the variants which have emerged from it in economics.

In practice, the three types of research are usually not easily separated. A given research endeavor/project may embody multiple types. Some Ph.D. dissertations, for example, make theoretical or other disciplinary contributions, then proceed on with a subject-matter application and even to some problem-solving prescriptions. Some believe that the discipline would be better served with somewhat more emphasis on empirical analytical validation of theory (Leontief, 1993; Caldwell, 1982; Eichner, 1986; O'Sullivan, 1987, p. 74; Blaug, 1980, ch. 15; Tomek, 1993) as well as the empirical estimation process in subject-matter and problem-solving research.

The position advocated here is that every type of research is relevant, because we need all types. It is quite possible, however, that the collective discipline can be out of balance among the types. If we have, as some maintain, placed too much

relative emphasis on disciplinary journal publishing compared to other types of outlets within the discipline, there are two basic forces that may have caused it. First, academic administrators (about two-thirds of Ph.D. economists are employed by academic institutions [Hansen, 1991, p. 1060]) are prone to recognize only journal publishing as evidence of research productivity, coupled with the propensity for economics journals to want to publish only research that is primarily disciplinary in its focus. This system of rewards and constraints has created a situation in which economists, particularly academicians, increasingly communicate only with other economists, and only about things that other users of our research may view as too esoteric to be relevant. Secondly, in their desire to be more scientifically rigorous in a limited positivistic sense, economists have become increasingly enthralled with mathematics, sometimes losing the reason for employing mathematics (to gain more precision in economic concepts) in the rush to express thoughts in mathematical terms and sometimes at the expense of empirical testing and verification. Among other problems (e.g., making assumptions about economic behavior or motivation merely because they facilitate mathematical derivation or deterministic solutions), this fascination may have led us further away from producing applied economic research that can be understood by potential users of our research.

On the other hand, a part of the problem is that most noneconomist users of our research wish for results to be intuitively obvious, with little need for them to understand the complexities of why results occur as they do. This is not possible with many of the complex phenomena that are the objects of economic analysis. A related problem is that very few of our corporate and public policy research users who are interested in applied economic research results understand the need for disciplinary research as the capital stock for applied research. Although solutions to this dilemma are elusive, we need to recognize it because in the final analysis it is our problem; we are the ones who must obtain the support for our discipline and the research within our discipline.

To synthesize how different philosophical positions have merged to affect economic thinking and economic research, Randall (1993, p. 54) states:

From the rationalists, we have learned that logical coherence is a highly desirable property of an argument, and we have some well-established principles of logic to guide us. From the empiricists we have learned to respect the evidence and to develop procedures to impose *ceteris paribus* and to come to terms with stochastic phenomena. The logical positivists taught us to respect the distinction between empirical and metaphysical propositions, just as the disputes that led to the dismantling of their school taught us that the distinction is not quite as simple as it might seem. From falsification, we learned to cherish the opportunities to conduct a definitive test of an interesting, refutable hypothesis, on the rather rare occasions such opportunities are presented to us.

A similar synthesis, stated another way (Johnson, 1997, pp. 262–263) is:

Economics has difficulty being either *just* logically positivistic or *only* normativistic. If economics is to define and locate optima, it requires both the value-free knowledge obtainable with a logically positivistic approach and the knowledge of values obtainable with a normative approach; this implies either (1) both a logically positivistic and normativistic orientation or (2) a pragmatic orientation.

EMPIRICISM IN RESEARCH METHODOLOGY

The influence of the philosophy of logical positivism fostered interest in measurement or quantification—use of data to test validity of theories and derive expected magnitudes of effects as a basis for policy recommendations. Logical empiricism may be viewed as going beyond logical positivism to subjecting the results of logical positivism to testing.³ There was an interest in quantification of economic phenomena with the classicists, but quantification remained very limited until the early twentieth century. The simultaneous evolution of several technologies and events has coincided in the last century to produce emphasis on empiricism—measurement and quantification—in economic matters.

The interest of governments in the industrialized countries in collecting and compiling data to produce many types of economic data and indicators has been associated with positivistic and pragmatic philosophies (a recognized need to know what it is and to use that knowledge in seeking solutions to perceived problems). This interest in producing social and economic data has both fostered the development of statistical methods, especially that branch of statistics called sampling, and been fostered by statistical methods, especially the estimation techniques branch of statistics. The emphasis on data, some of it dealing with factual information about values, has been an important element of empirical developments in economics. The development of the theory and methods of statistical analysis during the twentieth century has enabled economists to exploit those data.

As economists learned about statistics, its integration with mathematics and economic theory led to the development of econometrics. Econometrics represents both a subject-matter specialization and a necessary part of every economist's basic analytical training. Econometrics has its origins in the early 1900s, but it achieved "legitimacy" in the 1940s and 1950s, made more effective by the development of electronic computers beginning in the 1940s. Econometrics includes empirical data and measurements, making it both more positivistic and more pragmatic and thus more than merely the abstract logic of mathematical economics (see Johnson, 1986b, p. 84, for further discussion of this distinction).

The influence of applied areas of economics in the evolution of logical empiricism and the development of econometrics, as well as other empirical

quantitative techniques, has been considerable. Some excellent perspectives and interpretations of these matters are provided in Leontief (1993), Bonnen (1986), Breimyer (1991), Castle (1989), and McCloskey (1990a). In many instances in the evolution of econometric methods, the applied problem-solving focus of the applications led to the questions that identified the need for econometric concepts and theory. A corresponding influence was the role of the institutionalist perspective by the 1930s. The role of institutions was made more obvious by the expanding influence of the U.S. government in public policy matters. The empirical bent (including training in statistical methods) made the contingent of applied economists with sound theoretical knowledge suited for a leadership role in the development of econometrics.

Several leading economists, including Wassily Leontief (1982), a leader in quantitative methods in economics, have pleaded for more empiricism in economics. They look to measurement, reliance on data (objective evidence), and quantitative analysis as the vehicles to place economics on the same scientific footing as, for example, physics.

THE SCIENTIFIC APPROACH

The general process called the scientific approach,⁴ which some refer to as the "scientific method," is influenced by the philosophical views previously discussed. We will avoid characterizing the process as the scientific method because there is no single scientific method. There is, however, a central scientific methodology or scientific approach (the two terms may be used interchangeably in this context). It may be presumptuous to label it *the* scientific approach because it is comprised of many variations. The rationalization for doing so is for brevity; if the label is objectionable, substitute the term "a generalized manner of approaching research" or some such descriptive phrase.

The scientific approach may be characterized as having the following general steps: (1) identify the problem/issue/question, which may be disciplinary, subject-matter, and/or problem-solving in nature; (2) define the research objectives; (3) develop approaches for achieving objectives (which may include hypotheses of expected outcomes and/or alternative solutions); (4) conduct the analysis (obtain appropriate information and evaluate it, which may include testing of hypotheses); and (5) interpret the results and draw conclusions, including providing prescriptions, if appropriate. This general list of steps may vary with circumstances. However, if one views the above steps as a guideline for a general progression rather than as an inflexible, lockstep procedure, it is adaptable to most research situations.

The above list of steps is common across disciplines except for step 5. In some fields of research, particularly in the physical sciences, interpreting results and/or drawing conclusions is often omitted. Prescriptions are also less likely to be provided in disciplinary research. Interpretation of results—evaluating the

implications of rejecting hypotheses or not—is universally accepted as a part of the process in the social sciences and seems to be increasingly accepted in the physical sciences. Einstein (1991, p. 829) notes that the scientific approach, without step 5, can teach us nothing beyond how facts are related to and conditioned by each other, and intelligent thinking must play a role in the approach. Also, the Committee on Science, Engineering, and Public Policy (1995, p. 2) states that empirical evidence alone, without step 5, is not sufficient. The reader should note the similarities or overlap between this list of steps in the scientific approach and the diagram of steps in the research process in chapter 2 (see figure 2.2) and the relationship between research and science discussed in chapter 3.

The scientific approach is often presumed, especially by persons in the physical sciences, to be based entirely on positivistic philosophy. That is an indefensible position to most thoughtful positivistic, as well as normativistic and pragmatic, economists. Identifying a problem and deciding on research objectives contain elements of normativism and may well contain elements of pragmatism. Problem identification must be affected by individual and group perceptions: that is, what we perceive to constitute a problem. Personal characteristics of mind, individual and public values, and priorities are inherent in the process of problem identification, including disciplinary problems. Likewise, objectives—the identified set of specific goals that are related to the problem—are inherently normative to some extent. A decision must be made about what objectives are appropriate, given the problem. Both problem identification and objectives specification may have a pragmatic orientation. For example, a problem may focus on knowledge that is needed for prescriptions, with the corresponding objectives being designed to provide that knowledge and/or prescriptions. The approach will typically contain more prominent elements of pragmatism when it is focused on problem-solving or subject-matter research than on disciplinary research.

Steps 3 and 4, developing and conducting the analysis, data collection, and testing hypotheses, may be, but are not necessarily, positivistic. In any event, it is through these steps that disciplines are concerned with validation of concepts, as well as hypotheses about results. As has been noted, normativistic knowledge is usually involved in economics and the other social sciences. Recall that the division between logical positivism and objective normativism is somewhat nebulous. These steps, applied for many years in various forms of the classical experimental design in the laboratory and field sciences, are perceived by some to consist entirely of *statistical* data generation and hypothesis testing. As noted in chapter 3, hypotheses and hypothesis testing cannot necessarily be limited to statistical hypotheses. Hypotheses may also be prescriptive (remedial), as noted by Tweeten (1983).

The laboratory and field sciences base these steps of the process more heavily on positivistic philosophy than do the social sciences. The differences in applications of the steps is probably due more to the evolution of disciplines' perceived or defined roles than to any other single factor. The laboratory and field sciences

see their research process as producing reliable *data* (e.g., much of their attention is devoted primarily to experimental design to generate statistically valid numbers). The social sciences see their process more in terms of using data to understand *relationships* (e.g., their attention dwells on using numbers to generate conceptually and statistically valid relationships) and to address problems that require decisions.

This latter point is related to the differences among disciplines regarding the role of the last step, interpreting results. Persons whose philosophy is more positivistic may argue that interpretations introduce normativistic elements into the process. Economists concur with that observation but maintain that normative interpretation is a valuable addition if it is done with personal objectivity (neutrality). The data and/or relationships generated or estimated through the process do not necessarily "speak for themselves," as might be maintained by the positivist. They furnish knowledge or understanding, but it is dependent to a degree on the conditions under which it was generated. The scientist's clarification of those conditions can help avoid misinterpretation or misuse of the information. If the dangers of misinterpretation are greater when the scientific approach is applied to more complex relationships in the social sciences, this may explain the differing views among disciplines about appropriateness of interpretation.

In essence, differences in specific applications of the scientific approach revolve around different emphasis on (1) the type of problem (disciplinary, subject-matter, or problem-solving); (2) the nature and types of hypotheses, including explicit vs. implicit and quantitative vs. qualitative; and (3) the roles and functions of implications, conclusions, and/or prescriptions. These variations are not too surprising upon recognition of the different roles and functions in the larger picture of generating new, reliable knowledge of different types. The disagreements seem to stem from the lack of understanding or perspective. They may be unproductive. Different fields and types of research have reasons to place more or less emphasis on the various steps of the approach.

Deduction and Induction in the Scientific Approach

Except in the somewhat specific circumstances when the intent of research is totally on development of theoretical constructions, the approach in economics consists of an ongoing interfacing of deduction and induction. Even when the focus is on developing theory, a deductive process, the theory must eventually be evaluated for applicability or validated through empirical testing, an inductive process. Volumes have been written on deduction and induction and their uses and functions in science, including economics (see, e.g., Larrabee, 1964, pp. 65-91, 236-255; Caldwell, 1982, pp. 36-45; Johnson, 1986b, p. 77). Our treatment of deductive and inductive reasoning is focused on the ways they are used instead of philosophical debates about relative merits.

In deductive reasoning we start from premises—assumptions. Deductive reasoning holds that if individual premises are true and collectively complete and the

reasoning is correct, the conclusion is necessarily reliable. This enables us to organize premises into patterns that provide valid conclusions; however, the conclusions cannot exceed the logical content of the premises (Ghebremedhin and Tweeten, 1994, p. 9).⁵ The discipline of mathematics is based largely on deductive reasoning; it is tautological.⁶ The most direct application of deductive reasoning within the scientific approach may be in step 3, developing hypotheses. For example, in arriving at hypotheses regarding expected outcomes in economics, we arrive at those expectations through a process of theoretical logic. In disciplinary research the theoretical reasoning may be newly developed; it may entail new concepts, new conceptual elements added to existing theory, or conceptual modification of existing theory, all of which lead to hypothesized outcomes. In research with more applied elements the reasoning will involve the application of existing theory to the issue of concern, with the resulting hypothesis relating to the expected outcome in a specific situation. In the former case, the hypothesis is not only of a different nature, it is more central to the basic intent of the research. In the latter case, economists often neglect to state a formal hypothesis. This is particularly the case with econometric models, for example, and when we are dealing with qualitative hypotheses.

Induction is an empirical process of arriving at new generalities from observed realities (facts, data) and does not depend on previous knowledge (Gebremedhin and Tweeten, 1994, p. 10). Judgments about the generalities can be made based on the probability of making errors. In fact, Churchman (1948) presented statistical inference as a decision theory of science, not as a scientific method. Again, while the process of arriving at hypotheses is largely deductive, the testing of hypotheses, statistical or otherwise, is largely inductive.

As economists we receive formal training in the use of statistical induction. Statistical induction is the process of testing whether estimates of parameters are different from some specified quantity or whether estimates of relationships explain a statistically significant proportion of the variation in variables we are attempting to explain. For the statistical tests to be valid, the population and sample data distributions must meet specific conditions. When we test the statistical significance of, for example, the income coefficient for the demand for clothing, we are attempting to infer inductively something about the impact of income changes on clothing purchases.

Induction cannot constitute proof of a proposition because we have not examined (cannot examine) all possible evidence relevant to the case (Larrabee, 1964, p. 236). Induction is the process of reasoning from specific circumstances to generalized outcomes. We can show that the outcome is probable on the basis of available evidence that has been generated or obtained from reliable information. We cannot say that an outcome or relationship will always hold because it has held in every case that we have examined; the underlying causal structure supporting the outcome or relationship could change (Caldwell, 1982, pp. 40-41). But the outcome based on the evidence is probable, or even has a specified probability.

Although not as highly formalized, the fifth step in the scientific approach is nevertheless largely inductive. When we develop and examine implications of the findings, we are often extrapolating from the specific analysis to the more general applications or prescriptions. Implications may be in the form of policy or management outcomes/prescriptions or in the form of more basic understanding of phenomena or values. As in the more formal inductive process of quantitative hypothesis testing, the specific outcome is not necessarily of primary interest, but it *may* suggest outcomes in the general case. Examples of this in economics lie in the use of Marshall's (1964) representative firm as a vehicle to reason through the expected behavior of the suppliers in an industry. If firms are similar in their cost structures and technology adoption, an increase (decrease) in the price of the good being produced will elicit an output response by individual firms such that the industry supply curve can be viewed as the horizontal summation of the firms' marginal cost curves. This reasoning is sometimes used in empirical research, particularly when the industry is structurally competitive. By modeling the structure/behavior of a representative (i.e., average or typical) firm, the outcome for the industry may be estimated to be the outcome for the firm times the number of firms in the industry. These types of inferences require explanation and defense of the inductive reasoning by researchers on a case-by-case basis.

Identifying the research problem and objectives are both inductive and deductive, with the dominant logic depending in part on the point of departure. If one begins with a vague curiosity about something, the process is likely to be more deductive. Most of us have on occasion begun a research process with "I wonder why (or how, or if) . . .". When the process is initiated in this manner, one moves from that vague curiosity to more carefully framing the dimensions of the curiosity, then to specifying the intent of the curiosity into a perspective of the issue that lends itself to development of new information, then to research objectives. This is a deductive process of moving from the general (curiosity) to the specific (research objectives).

Alternatively, if one starts with a "felt difficulty," which might take the specific form of a general objective or a researchable problem after being formalized, the thinking process may assume more of an inductive character. If one starts with a specific question or issue ("I would like to determine [or know, or find out] . . ."), then the question of why it is important to get an answer follows in order to justify the objective. This thinking or questioning process moves from the specific (a general objective) to the general (an explanation or understanding of the problem) and is inductive.

Both approaches to problem description are valid and neither is necessarily superior. The deductive approach to identification of problems and objectives may be more common with disciplinary research, while the inductive approach is more the case with applied research. More will be discussed on this matter in chapter 6.

Obtaining data in empirical economic research depends on a deductive reasoning process. Knowing what data are needed is preceded by a process of con-

ceptualizing the relevant variables and often a step involving "model building." Thus, we attempt to deduce the relevant variables on which we need to obtain data. The process of obtaining the data may involve induction if statistical procedures are a part of the process.

Francis Bacon viewed research as a purely inductive process (empiricism) in which we start with facts (what our senses reveal) and proceed "upward" to axioms, then laws (Back, no date). However, the scientific approach integrates the inductive and deductive methods of reasoning. Deduction provides the necessary implications of premises, which may be general (laws, axioms, principles) or specific (factual), while induction examines the validity or applicability of the premises. In isolation, either approach leaves gaps in our understanding. Deductive reasoning can organize what is already known and deduce new relationships, but is not sufficient as a source of new knowledge; inductive reasoning fails to use prior knowledge, (including theory), so it is inefficient.

Deduction (i.e., theorizing) alone is fallible in studying the real world because it lacks definitive means of evaluating whether the premises (assumptions) of the "ideal types" match the situation being analyzed. Induction alone (i.e., observing, empirical testing) is fallible because there is always a probability of error (Ladd, 1991, pp. 4, 5). Put another way, "no theoretical question is answerable without some answers to questions of fact, and vice versa" (Back, no date). Scientific inquiry must rely on the use of both deduction and induction in a constant interaction with one another; we call it the scientific approach.

SUMMARY

Three major philosophies—positivism, normativism, and pragmatism—form the philosophical basis for research methodology in economics. Each philosophy affects economic thinking and economic research, yet no individual philosophy is dominant and no single philosophy sufficiently encompasses the range of economic research. Economics blends the philosophies, as do other disciplines, with the specific mixture varying with the type of research, even the interests of researchers.

Logical positivism as a philosophy related to research emphasizes determination of value-free information and knowledge of values about phenomena and its logical extensions, and its focus is on determining "what is." Logical positivism has influenced economics during the twentieth century, with its impetus facilitated by empirical data availability and inductive techniques and technology. The *relative* influence of logical positivism may have declined since the last one-third of the twentieth century. Objective normativism as a philosophy is imbedded in the foundations of economics. We study how and why people, groups, and societies value things. Economists view objective knowledge of values as being much like positivistic knowledge of physical entities in regard to usefulness, validity, and rigor of the knowledge. Economists also embrace prescriptive normativistic knowledge, conditional on it being personally objective.

Pragmatism as a philosophy in economics is revealed in the emphasis on relevance—our attempts to produce knowledge that is useful in addressing contemporary problems. It does not distinguish between value-free knowledge and knowledge of values because both are interdependent for addressing practical problems. Pragmatism is tempered by overriding considerations for theoretical and analytical rigor, but the identifiable interest in “workability” in our economic research is based on pragmatic beliefs.

The scientific approach—identifying the problem/question/issue, defining objectives, developing hypotheses or alternative solutions, designing research procedures, analyzing information, and interpreting/concluding/prescribing—adapts elements of the three philosophies. Each philosophy may assume different roles and importance in the various components in the scientific approach. Deduction and induction, as forms of reasoning, also have critical roles in the scientific approach. While both are important, with each having functions in different components of the scientific approach, their interaction in the research process allows us to reach implications and conclusions that neither form of reasoning can achieve on its own.

RECOMMENDED READING

Johnson. Chapters 4–9, *Research Methodology for Economists*.

Ladd. “Thoughts on Building an Academic Career,” 3–6.

McCloskey. *If You’re So Smart*, pp. 1–9.

McCloskey. “The Rhetoric of Economics.”

SUGGESTED EXERCISES

1. Choose a research article from a recent issue of an economics journal. Write a description of how the research reported on was influenced by the philosophies of positivism, normativism, and pragmatism. Comment on how the philosophies have been blended.

2. Select any recently published report of research. Provide a written description of how deductive and inductive logic were used in conducting the research.

3. Based on your own reading (knowledge of literature) or research experience, cite an example of research that demonstrates the pragmatic interdependence of value-free knowledge and knowledge of values.

NOTES

1. The ideas offered on this distinction were initially provided to the author by George W. Ladd.
2. This should not be confused with conditional normativism, although they are related to one another. *Conditional normativism* takes an individual’s or group’s values as given without judging

them by subjective criteria; conditional normativism is conditioned on assumptions. *Objective normativism* holds that statements or knowledge about values (good and bad) may be as objective as statements or knowledge of physical things; "both are mental and products of the mind" and both require faith and interpretation (Johnson, 1986b, p. 58). Both objective and conditional normativism prescribe actions or solutions that achieve goals defined by the individual's or group's values.

3. This perspective was provided by Warren Samuels.
4. This refers to a general manner of going about obtaining new knowledge or understanding.
5. Sometimes the premises are true by definition. When this is the case, the conclusions drawn from them are called *tautologies*.
6. Logic and mathematics are tautological and nonempirical (not testable) and are the only disciplines where certain answers are possible (Simkin, 1993, p. 16). They also have nothing to say about "reality."